

FOUR FACTORIZATIONS OF K_{35} REALIZING $\beta \in \{1, 2, 4, 6\}$ IN THE HAMILTON–WATERLOO PROBLEM $\text{HWP}(35; 5, 7)$

ABSTRACT. We print four factorizations of K_{35} into seventeen two-factors, realizing $(\alpha, \beta) = (16, 1), (15, 2), (13, 4)$ and $(11, 6)$ in the Hamilton–Waterloo problem $\text{HWP}(35; 5, 7)$. Each object is given in full as a list of cycles, and checking one is a finite computation on that list. Burgess, Danziger and Traetta determined $\text{HWP}(mn; m, n)$ for odd $n \geq m \geq 3$ up to an explicit list of possible exceptions, whose clause at $(m, n) = (5, 7)$ names the cells with $\beta \in \{1, 2, 4, 6\}$ and the quadruple $(\alpha, \beta) = (9, 8)$. Section 1 paraphrases, from its e-print, a theorem of Wang, Lu and Cao covering the fourteen pairs at $v = 35$ with $\beta \notin \{1, 2, 4, 6\}$; granting that statement, which is neither displayed nor verified here, these four factorizations account for the remaining four of the eighteen pairs with $\alpha, \beta \geq 0$ and $\alpha + \beta = 17$. No priority is claimed for the four cells.

1. THE PROBLEM AND THE TWO QUOTED THEOREMS

For $v \geq 3$ and $3 \leq m, n \leq v$, the Hamilton–Waterloo problem $\text{HWP}(v; m, n)$ asks for which pairs (α, β) the edge set of K_v decomposes into α two-factors all of whose cycles have length m and β two-factors all of whose cycles have length n . We write $(\alpha, \beta) \in \text{HWP}(v; m, n)$ when such a decomposition exists. For odd v a two-factorization of K_v has exactly $(v - 1)/2$ factors, so $\alpha + \beta = (v - 1)/2$ is forced; and $m \mid v$ is forced when $\alpha \geq 1$, $n \mid v$ when $\beta \geq 1$. At $(m, n, v) = (5, 7, 35)$ both divisibilities hold, so the only necessary condition in force below is $\alpha + \beta = 17$.

Burgess, Danziger and Traetta [1] proved, for odd $n \geq m \geq 3$ and $v = mn$, that $(\alpha, \beta) \in \text{HWP}(mn; m, n)$ if and only if $\alpha, \beta \geq 0$ and $\alpha + \beta = (mn - 1)/2$, *except possibly* when

$$\beta \in \left[1, \frac{n-3}{2}\right] \cup \left\{\frac{n+1}{2}, \frac{n+5}{2}\right\}, \quad \text{or} \quad (m, \alpha) \in \{(3, 2), (3, 4)\},$$

or (m, n, α, β) is one of seven listed quadruples. This statement, and that of [2] paraphrased below, are read off the e-prints of those papers rather than from the typeset journal pages; neither is reproduced verbatim here, and nothing below verifies either of them.

At $(m, n) = (5, 7)$ we have $v = 35$ and $\alpha + \beta = 17$. Here $[1, (n - 3)/2] = \{1, 2\}$ and $\{(n + 1)/2, (n + 5)/2\} = \{4, 6\}$, the clause $(m, \alpha) \in \{(3, 2), (3, 4)\}$ is inert because $m = 5$, and exactly one of the seven listed quadruples has

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$(m, n) = (5, 7)$, namely $(5, 7, 9, 8)$. So the exception clause of [1] names five cells at $v = 35$, those with $\beta \in \{1, 2, 4, 6, 8\}$.

On our reading of the e-print of [2], Wang, Lu and Cao prove $(9, 8) \in \text{HWP}(35; 5, 7)$ outright, and their general theorem for odd orders excepts, at $v = 35$, only $\beta \in \{1, 2, 4, 6\}$; that is, it gives the fourteen pairs with $\beta \notin \{1, 2, 4, 6\}$, the cell $(9, 8)$ among them. We use this only as quoted, in Corollary 2. The four factorizations below realize the pairs with $\beta \in \{1, 2, 4, 6\}$; whether those four pairs had been realized before is not settled here.

2. TWO PRESCRIBED AUTOMORPHISMS

Take $V(K_{35}) = \{0, 1, \dots, 34\}$. We use two permutations of V , each of prime order, and each acting on V in blocks of consecutive integers.

The $p = 7$ lane. Write $u = 7c + x$ with $c \in \{0, \dots, 4\}$ and $x \in \mathbb{Z}_7$, and set

$$\rho(7c + x) = 7c + ((x + 1) \bmod 7).$$

Thus ρ has order 7 and fixes each of the five *classes*

$$\{0, \dots, 6\}, \{7, \dots, 13\}, \dots, \{28, \dots, 34\}$$

setwise.

The $p = 5$ lane. Write $u = 5j + y$ with $j \in \{0, \dots, 6\}$ and $y \in \mathbb{Z}_5$, and set

$$\sigma(5j + y) = 5j + ((y + 1) \bmod 5),$$

of order 5, fixing each of the seven classes

$$\{0, \dots, 4\}, \{5, \dots, 9\}, \dots, \{30, \dots, 34\}$$

setwise.

Both actions on $E(K_{35})$ are semiregular, so all edge orbits have size p . For ρ there are 85 orbits: fifteen *intra-class* orbits, the three circulants $\text{Cay}(\mathbb{Z}_7, \{\pm d\})$, $d = 1, 2, 3$, inside each of the five classes, and seventy *inter-class* orbits, one for each of the ten pairs of classes and each of the seven shifts; and $85 \times 7 = 595 = \binom{35}{2}$. For σ there are 119 orbits, 14 intra-class and 105 inter-class, and $119 \times 5 = 595$. An intra-class orbit is a p -cycle on its class; an inter-class orbit is a perfect matching between its two classes.

A two-factor of K_{35} fixed setwise by the lane map is called *invariant*; it is a union of $35/p$ edge orbits and counts as one factor. A two-factor F whose images $F, \rho F, \dots, \rho^{p-1} F$ (respectively $F, \sigma F, \dots$) are pairwise edge-disjoint is called a *base* object; it stands for p factors and uses up 35 edge orbits. This is the whole bookkeeping: a certificate is a short list of objects whose expansion must have 17 factors using each of the 595 edges exactly once.

3. THE FOUR CELLS

Theorem 1. $(16, 1)$, $(15, 2)$, $(13, 4)$ and $(11, 6)$ all lie in $\text{HWP}(35; 5, 7)$.

Proof. Section 4 prints, for each of the four cells, a list of objects in one of the two lanes. Expanding each base object into its p translates gives in every case 17 two-factors; the verification of Section 5 confirms that each factor is a C_5 - or C_7 -factor as declared, that the 17 factors are pairwise edge-disjoint with union all of $E(K_{35})$, and that the number of C_5 - and C_7 -factors is the claimed pair (α, β) . $\square$

Corollary 2. *Suppose the theorem of [2] gives the fourteen pairs (α, β) with $\alpha, \beta \geq 0$, $\alpha + \beta = 17$ and $\beta \notin \{1, 2, 4, 6\}$, as its e-print states and as paraphrased in Section 1. Then*

$$\text{HWP}(35; 5, 7) = \{(\alpha, \beta) : \alpha, \beta \geq 0, \alpha + \beta = 17\};$$

that is, all eighteen admissible pairs are realizable.

Proof. $\alpha + \beta = 17$ is necessary because 35 is odd. The fourteen pairs with $\beta \notin \{1, 2, 4, 6\}$ are the hypothesis, and the four with $\beta \in \{1, 2, 4, 6\}$ are Theorem 1. $\square$

The hypothesis of Corollary 2 is quoted from an e-print and is neither displayed nor verified here; only the four cells of Theorem 1 are established below.

4. THE CERTIFICATES

Each object is printed as its decomposition into cycles: a C_5 -factor as seven 5-cycles and a C_7 -factor as five 7-cycles, together spanning $\{0, \dots, 34\}$. An object marked *base* stands for itself and its $p-1$ translates under the lane map; an object marked *invariant* stands for one factor and is fixed setwise by that map.

$(\alpha, \beta) = (16, 1)$, **lane** $p = 7$. G_1 : a C_7 -factor, ρ -invariant.

$$\begin{aligned} & (0, 2, 4, 6, 1, 3, 5)(7, 9, 11, 13, 8, 10, 12) \\ & (14, 16, 18, 20, 15, 17, 19)(21, 24, 27, 23, 26, 22, 25) \\ & (28, 31, 34, 30, 33, 29, 32) \end{aligned}$$

G_2 : a C_5 -factor, *base*.

$$\begin{aligned} & (0, 12, 11, 23, 31)(1, 22, 18, 28, 33)(2, 8, 4, 5, 19) \\ & (3, 10, 13, 26, 16)(6, 7, 24, 30, 25)(9, 15, 27, 21, 32) \\ & (14, 17, 29, 20, 34) \end{aligned}$$

G_3 : a C_5 -factor, *base*.

$$\begin{aligned} & (0, 22, 2, 5, 34)(1, 10, 28, 8, 26)(3, 13, 21, 30, 29) \\ & (4, 27, 25, 17, 32)(6, 14, 15, 24, 18)(7, 31, 20, 11, 33) \\ & (9, 19, 12, 16, 23) \end{aligned}$$

G_4 : a C_5 -factor, ρ -invariant.

$$(0, 16, 11, 27, 30)(1, 17, 12, 21, 31)(2, 18, 13, 22, 32) \\ (3, 19, 7, 23, 33)(4, 20, 8, 24, 34)(5, 14, 9, 25, 28) \\ (6, 15, 10, 26, 29)$$

G_5 : a C_5 -factor, ρ -invariant.

$$(0, 18, 10, 31, 24)(1, 19, 11, 32, 25)(2, 20, 12, 33, 26) \\ (3, 14, 13, 34, 27)(4, 15, 7, 28, 21)(5, 16, 8, 29, 22) \\ (6, 17, 9, 30, 23)$$

$(\alpha, \beta) = (15, 2)$, **lane** $p = 7$. G_1 : a C_5 -factor, *base*.

$$(0, 23, 31, 21, 28)(1, 10, 20, 15, 32)(2, 6, 17, 33, 12) \\ (3, 11, 8, 4, 25)(5, 18, 27, 16, 30)(7, 19, 26, 9, 22) \\ (13, 24, 14, 34, 29)$$

G_2 : a C_5 -factor, *base*.

$$(0, 17, 1, 20, 26)(2, 8, 24, 10, 16)(3, 18, 9, 4, 21) \\ (5, 12, 32, 7, 31)(6, 28, 13, 14, 33)(11, 15, 27, 19, 30) \\ (22, 23, 29, 25, 34)$$

G_3 : a C_5 -factor, ρ -invariant.

$$(0, 27, 8, 15, 30)(1, 21, 9, 16, 31)(2, 22, 10, 17, 32) \\ (3, 23, 11, 18, 33)(4, 24, 12, 19, 34)(5, 25, 13, 20, 28) \\ (6, 26, 7, 14, 29)$$

G_4 : a C_7 -factor, ρ -invariant.

$$(0, 1, 2, 3, 4, 5, 6)(7, 9, 11, 13, 8, 10, 12) \\ (14, 15, 16, 17, 18, 19, 20)(21, 24, 27, 23, 26, 22, 25) \\ (28, 31, 34, 30, 33, 29, 32)$$

G_5 : a C_7 -factor, ρ -invariant.

$$(0, 2, 4, 6, 1, 3, 5)(7, 8, 9, 10, 11, 12, 13) \\ (14, 17, 20, 16, 19, 15, 18)(21, 23, 25, 27, 22, 24, 26) \\ (28, 29, 30, 31, 32, 33, 34)$$

$(\alpha, \beta) = (13, 4)$, **lane** $p = 5$. G_1 : a C_5 -factor, *base*.

$$(0, 21, 9, 5, 33)(1, 15, 16, 27, 26)(2, 25, 24, 13, 29) \\ (3, 18, 11, 10, 20)(4, 28, 19, 6, 31)(7, 14, 23, 22, 17) \\ (8, 30, 32, 12, 34)$$

G_2 : a C_5 -factor, *base*.

$$\begin{aligned}
&(0, 20, 29, 34, 26)(1, 10, 27, 31, 30)(2, 18, 7, 25, 32) \\
&(3, 11, 5, 13, 16)(4, 9, 24, 6, 22)(8, 15, 14, 12, 17) \\
&(19, 23, 28, 21, 33)
\end{aligned}$$

G_3 : a C_5 -factor, σ -invariant.

$$\begin{aligned}
&(0, 1, 2, 3, 4)(5, 10, 19, 32, 26)(6, 11, 15, 33, 27) \\
&(7, 12, 16, 34, 28)(8, 13, 17, 30, 29)(9, 14, 18, 31, 25) \\
&(20, 22, 24, 21, 23)
\end{aligned}$$

G_4 : a C_5 -factor, σ -invariant.

$$\begin{aligned}
&(0, 2, 4, 1, 3)(5, 7, 9, 6, 8)(10, 29, 21, 19, 31) \\
&(11, 25, 22, 15, 32)(12, 26, 23, 16, 33)(13, 27, 24, 17, 34) \\
&(14, 28, 20, 18, 30)
\end{aligned}$$

G_5 : a C_5 -factor, σ -invariant.

$$\begin{aligned}
&(0, 6, 10, 33, 24)(1, 7, 11, 34, 20)(2, 8, 12, 30, 21) \\
&(3, 9, 13, 31, 22)(4, 5, 14, 32, 23)(15, 17, 19, 16, 18) \\
&(25, 27, 29, 26, 28)
\end{aligned}$$

G_6 : a C_7 -factor, σ -invariant.

$$\begin{aligned}
&(0, 7, 26, 19, 22, 30, 11)(1, 8, 27, 15, 23, 31, 12) \\
&(2, 9, 28, 16, 24, 32, 13)(3, 5, 29, 17, 20, 33, 14) \\
&(4, 6, 25, 18, 21, 34, 10)
\end{aligned}$$

G_7 : a C_7 -factor, σ -invariant.

$$\begin{aligned}
&(0, 8, 32, 21, 15, 25, 12)(1, 9, 33, 22, 16, 26, 13) \\
&(2, 5, 34, 23, 17, 27, 14)(3, 6, 30, 24, 18, 28, 10) \\
&(4, 7, 31, 20, 19, 29, 11)
\end{aligned}$$

G_8 : a C_7 -factor, σ -invariant.

$$\begin{aligned}
&(0, 9, 26, 11, 23, 33, 17)(1, 5, 27, 12, 24, 34, 18) \\
&(2, 6, 28, 13, 20, 30, 19)(3, 7, 29, 14, 21, 31, 15) \\
&(4, 8, 25, 10, 22, 32, 16)
\end{aligned}$$

G_9 : a C_7 -factor, σ -invariant.

$$\begin{aligned}
&(0, 10, 23, 9, 29, 16, 31)(1, 11, 24, 5, 25, 17, 32) \\
&(2, 12, 20, 6, 26, 18, 33)(3, 13, 21, 7, 27, 19, 34) \\
&(4, 14, 22, 8, 28, 15, 30)
\end{aligned}$$

$(\alpha, \beta) = (11, 6)$, **lane** $p = 5$. G_1 : a C_5 -factor, *base*.

$$\begin{aligned} & (0, 1, 4, 16, 15)(2, 5, 7, 24, 6)(3, 20, 18, 33, 23) \\ & (8, 9, 27, 29, 22)(10, 11, 13, 34, 21)(12, 17, 30, 31, 19) \\ & (14, 25, 26, 32, 28) \end{aligned}$$

G_2 : a C_5 -factor, *base*.

$$\begin{aligned} & (0, 19, 26, 20, 34)(1, 32, 2, 9, 33)(3, 12, 4, 10, 24) \\ & (5, 11, 8, 18, 14)(6, 22, 7, 28, 31)(13, 21, 23, 30, 25) \\ & (15, 17, 27, 16, 29) \end{aligned}$$

G_3 : a C_5 -factor, σ -invariant.

$$\begin{aligned} & (0, 5, 16, 13, 28)(1, 6, 17, 14, 29)(2, 7, 18, 10, 25) \\ & (3, 8, 19, 11, 26)(4, 9, 15, 12, 27)(20, 21, 22, 23, 24) \\ & (30, 32, 34, 31, 33) \end{aligned}$$

G_4 : a C_7 -factor, σ -invariant.

$$\begin{aligned} & (0, 6, 19, 30, 12, 24, 29)(1, 7, 15, 31, 13, 20, 25) \\ & (2, 8, 16, 32, 14, 21, 26)(3, 9, 17, 33, 10, 22, 27) \\ & (4, 5, 18, 34, 11, 23, 28) \end{aligned}$$

G_5 : a C_7 -factor, σ -invariant.

$$\begin{aligned} & (0, 10, 8, 27, 24, 19, 33)(1, 11, 9, 28, 20, 15, 34) \\ & (2, 12, 5, 29, 21, 16, 30)(3, 13, 6, 25, 22, 17, 31) \\ & (4, 14, 7, 26, 23, 18, 32) \end{aligned}$$

G_6 : a C_7 -factor, σ -invariant.

$$\begin{aligned} & (0, 12, 32, 9, 29, 20, 16)(1, 13, 33, 5, 25, 21, 17) \\ & (2, 14, 34, 6, 26, 22, 18)(3, 10, 30, 7, 27, 23, 19) \\ & (4, 11, 31, 8, 28, 24, 15) \end{aligned}$$

G_7 : a C_7 -factor, σ -invariant.

$$\begin{aligned} & (0, 18, 21, 32, 13, 8, 25)(1, 19, 22, 33, 14, 9, 26) \\ & (2, 15, 23, 34, 10, 5, 27)(3, 16, 24, 30, 11, 6, 28) \\ & (4, 17, 20, 31, 12, 7, 29) \end{aligned}$$

G_8 : a C_7 -factor, σ -invariant.

$$\begin{aligned} & (0, 23, 17, 5, 31, 14, 27)(1, 24, 18, 6, 32, 10, 28) \\ & (2, 20, 19, 7, 33, 11, 29)(3, 21, 15, 8, 34, 12, 25) \\ & (4, 22, 16, 9, 30, 13, 26) \end{aligned}$$

G_9 : a C_7 -factor, σ -invariant.

$$\begin{aligned}
&(0, 24, 14, 15, 6, 33, 26)(1, 20, 10, 16, 7, 34, 27) \\
&(2, 21, 11, 17, 8, 30, 28)(3, 22, 12, 18, 9, 31, 29) \\
&(4, 23, 13, 19, 5, 32, 25)
\end{aligned}$$

5. CHECKING A CERTIFICATE

Fix one of the four lists. Checking it is finite and needs nothing but the printed cycles.

- (1) Each printed object is a two-factor: its cycles are vertex-disjoint, they exhaust $\{0, \dots, 34\}$, and all have the declared length, so the object has 35 edges and the declared cycle type.
- (2) Each object marked invariant is fixed setwise by the lane map: applying ρ (or σ) to every printed edge returns the same edge set.
- (3) Each base object F has the p images $F, \rho F, \dots, \rho^{p-1} F$ pairwise distinct, with $\rho^p F = F$ (respectively for σ); these p factors have the same cycle type as F , the lane map being an automorphism of K_{35} .
- (4) The expansion has 17 factors, and their $17 \times 35 = 595$ edges are pairwise distinct and therefore exhaust $E(K_{35})$, whose size is $\binom{35}{2} = 595$. Equivalently, the union equals $E(K_{35})$ as a set.
- (5) Counting factors by type gives the claimed (α, β) , and $\alpha + \beta = 17$.

Step 4 is the only laborious one. It is where the orbit bookkeeping of Section 2 helps: the edges of an invariant factor form $35/p$ complete edge orbits and those of a base object 35 complete orbits, so it suffices to name the orbits used and check that the 85 (respectively 119) orbits are each named once. A machine check of all five steps for all four cells, in exact integer arithmetic, accompanies this note (`verify.py`, 66 checks, all passing); it reads the cycle lists exactly as printed above. It checks the five steps only: the hypothesis of Corollary 2 is quoted to it as a list of fourteen values of β , so the program tests the arithmetic of assembling the spectrum and not that hypothesis.

6. SCOPE

Only the instance $(m, n) = (5, 7)$, $v = mn = 35$, is treated; the other cells of the exception clause of [1] are untouched. The cell $(9, 8)$ is not ours: it is due to [2], and it is not printed here. We make no claim about uniqueness, enumeration, classification, minimality or nonexistence.

We make no priority claim either. What the four objects establish is that the four pairs are realizable, which is a statement about the pairs and not about the literature: it is not asserted here that their realizability was previously unrecorded, and no literature check settling that is reported. Falling outside the positive range of the two theorems cited above is a fact about those two theorems, not evidence that no other source realizes these cells.

REFERENCES

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